Experiment - 2

CISCO Packet tracer

Aim:

To study the Packet tracer tool Installation and user Interface.

Description:

This experiment introduced cisco packet tracer simulation tool and its user Interface for designing and simulation computer networks. It also covers the basics networking device such as hubs and switch. A hub broadcasts data to all connected devices whereas a switch forward data only to the intended destination using mac address making communication faster to more efficient.

Procedure:

1) Install and launch CISCO Packet Traces
2) Observe the user interface Including the device toolbar and workspace
3) Drag and place networking device such as PCs, hubs and switches onto the workspace.
4) Connect the device using the appropriate cables.
5) assign IP address to the PCs
6) save the network topology and record observation.

Observation:

a) Behaviour of switch and hub in forwarding packets

Hub: A hub forward the recieved data to all connected device without checking the destination address

Switch: A switch forwards the recieved data only to the destination device by using MAC address, making communication faster and more secure.

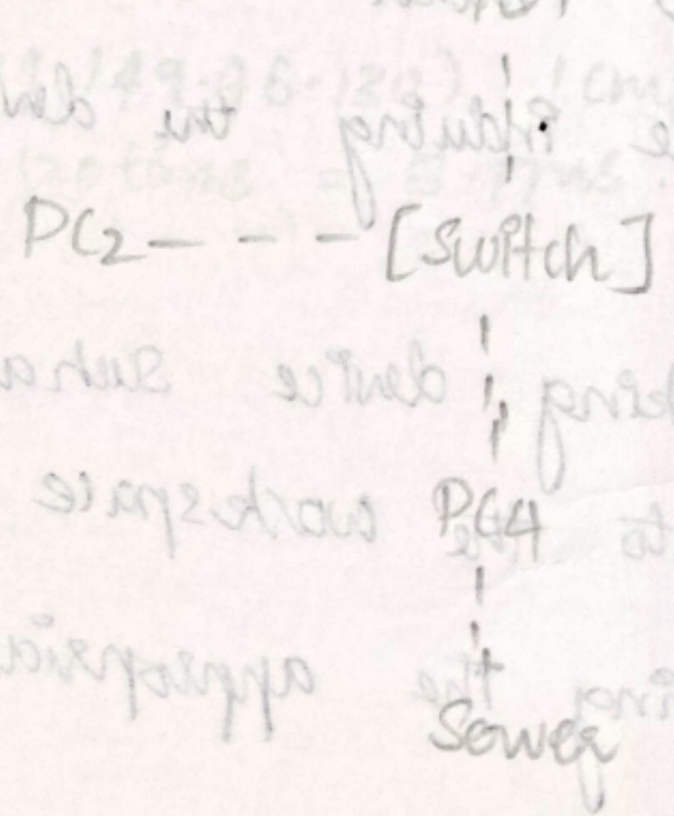
b) Network topology implements in the college:

Most colleges use a star topology where all computer are connected to central switch.

The hub broadcast packet to all connected device.

The switch forwards packet only to the intended destination using mac addresses.

diagram:



Result: The Cisco Packet Tracker environment was successfully explored. The working of hubs and switch was studied and verified.